Team Report Summary

Final Solution, Methods and Key Evidence

1. Scenario: Team-05 traced one communication, opening a secure website (HTTPS), from the client browser (192.168.1.10) to the web server (93.184.216.34) and back.
2. Solution: We delivered a 14-slide deck (12 content slides) and a recorded group presentation. We also produced an original packet journey diagram, a protocol analysis table, a TCP vs UDP comparison and a troubleshooting case.
3. Method: We followed the data layer by layer through the four TCP/IP layers: Application (data), Transport (segment), Internet (packet) and Network Access (frame). We mapped these to OSI Layers 5–7, 4, 3 and 1–2.
4. Application layer: DNS resolves the domain over UDP/53. The browser then builds an HTTP GET request, and TLS 1.3 encrypts it to form HTTPS.
5. Transport and Internet layers: The TCP three-way handshake (SYN, SYN-ACK, ACK) opens the session on port 443, with source port 51342. IP adds the source and destination addresses, and routers forward using only the destination IP.
6. Network Access layer: ARP resolves the gateway's MAC address, and Ethernet frames carry source and destination MACs. The MACs are rewritten at every hop, while the IP addresses stay constant.
7. Encapsulation and decapsulation: The sender adds headers in the order data, segment, packet, frame. The receiver strips them in reverse until the server is left with the original HTTP request, and the response returns by the same process.
8. Key evidence (diagram): The original packet journey diagram shows the sender, switch, gateway, ISP/internet, destination LAN and server, with the encapsulation and decapsulation stacks labelled.
9. Key evidence (tables): The protocol analysis table covers DNS, HTTPS, TCP, IP and Ethernet/Wi-Fi, with layer, PDU, addressing and purpose. The TCP vs UDP table compares connection, reliability, ordering, flow control and overhead.
10. Transport choice: TCP suits HTTPS and email (SMTP), where data must arrive complete and in order. UDP suits VoIP and DNS lookups, where speed matters more than perfect delivery.
11. Troubleshooting evidence: The failure was an expired certificate (`NET::ERR_CERT_DATE_INVALID`). A layered check using `ip link`, `ping`, `nslookup`, `nc`/`telnet` on port 443 and `openssl s_client` passed the first four layers and found the certificate expired.
12. Fix and references: We re-issued the certificate through a trusted CA, installed the full chain, reloaded the service and re-verified with OpenSSL. Sources: IETF RFC 8446, IETF RFC 793, Cisco Networking Academy (CCNA) and ICDFA Module 03.

AI Use Declaration

13. Declaration: In line with the course requirement, the team declares that AI assistance (Claude, by Anthropic) was used to help prepare the team project brief, the presenter scripts and speaker notes, this summary, and the GitHub README and report.
14. Purpose of use: AI was used as a support tool for structuring content, drafting wording and organising the material. The project scenario, the packet journey diagram, the failure case and the final slides remain the team's own work.

15. Verification: Every technical detail produced with AI help was reviewed by the team against the course material and the listed references (IETF RFC 8446, IETF RFC 793, Cisco Networking Academy and ICDFA Module 03) before submission.
16. Responsible use: AI output was treated as a draft, not a final answer. The team corrected, adapted and approved all content, and no AI-generated text was submitted without human review.
17. Accountability: Each member remains responsible for understanding and explaining the content. Every member presented in the recording and submitted an individual contribution declaration through the portal.